FOURIE VAN ROOYEN

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EDUCATION

The University of Alabama — Tuscaloosa, AL **May 2026**

Bachelor of Science in Aerospace Engineering | Major GPA: 3.0 | Dean's List: Fall 2025

Relevant Coursework: Solid Mechanics, Aerospace Structures, High-Speed Aerodynamics, Propulsion, Thermodynamics

RELEVANT SKILLS

Programming / Software

SolidWorks CAD, CATIA, ANSYS (CFD Simulations), MATLAB, Python, Delphi 10, Jira, CREO

Hardware / Technical

CNC Mill, CNC Lathe, 3D Printing, GD&T (ASME Y14.5-2018)

WORK EXPERIENCE

Mechanical Engineer I, Tactical Radars and Effectors — June 2026 – Present

Raytheon (RTX) — Tucson, AZ

- Support mechanical design and integration of the aft-section assembly for the Coyote counter-UAS effector platform, including the jet engine interface, fuel bladder assembly, and metal container structure.
- Develop and revise CAD models and technical drawings per GD&T (ASME Y14.5-2018) standards for jet engine mounting, fuel bladder, and metal container hardware to support manufacturability and tolerance stack-up.
- Collaborate with cross-functional engineering teams across structures, integration, and manufacturing to support design reviews and production readiness on the Tactical Radars and Effectors program.

Lab Technician — August 2025 – May 2026

The Cube — Tuscaloosa, AL

- Operate, maintain, and troubleshoot a fleet of advanced manufacturing equipment including FDM/FFF 3D printers, resin printers, waterjet cutting systems, and laser engravers to support academic, research, and student-led projects.
- Execute 3D printing using engineering-grade materials such as Nylon, Polycarbonate (PC), carbon-fiber–reinforced filaments, and specialty composites, optimizing print parameters for strength, surface finish, and dimensional accuracy.
- Support resin printing workflows, including part orientation, support strategy, washing, UV curing, and post-processing for high-resolution components.
- Assist students and faculty with CAD file preparation and slicing, ensuring manufacturability across additive and subtractive processes.

Ranch Hand — May 2022 – January 2024

Trinity Farm

- Provided daily care for horses including feeding, grooming, and medical assistance.
- Maintained farm infrastructure through mowing, fence repairs, and machinery troubleshooting.
- Operated and maintained various farm equipment and vehicles.
- Developed adaptability and critical thinking skills under varying conditions.

Tire Technician — August 2021 – February 2022

National Tire and Battery

- Performed routine maintenance: oil changes, tire repairs, battery installations, brake replacements.
- Assisted mechanics with advanced repairs and diagnostics.
- Provided customer service and explained repair needs to customers.
- Maintained a clean and organized work environment.

PROJECTS

Project ORCA — Fixed-Wing eVTOL Tiltrotor UAV (Capstone Project) — August 2025 – May 2026

Chief Engineer — The University of Alabama, Tuscaloosa, AL

- Led system-level design of a fixed-wing eVTOL tiltrotor reconnaissance UAV through a full NASA NPR 7123.1D design lifecycle, achieving stable VTOL hover and a successful VTOL→cruise→VTOL transition in flight testing.

- Executed full 3-D ANSYS Fluent CFD simulation at cruise (V_∞ = 17 m/s), ANSYS Mechanical FEA (FoS ≥ 2.0 on primary structure), and transient thermal analysis; validated aerodynamic L/D ≈ 10.5–11.2 and zero structural yield under flight loads.
- Configured and flight-validated ArduPlane QuadPlane firmware on a Matek F405-Wing V2 flight controller with DSHOT600 ESCs and a PCA9685 I^2C tilt-servo expander; resolved TIM8 timer conflict enabling simultaneous DSHOT and PWM operation.
- Analyzed ArduPilot DataFlash flight logs to identify root causes of two in-flight anomalies (EKF3 altitude divergence, UV-induced PLA structural softening); produced a 38-page Post-Flight Analysis Report documenting anomalies, corrective actions, and lessons learned.
- Conducted experimental crush testing on 3D-printed structures, identifying 7% gyroid infill as optimal strength-to-weight configuration.
- Oversaw cross-subsystem integration (aerodynamics, structures, propulsion, avionics, and manufacturing), resolving CG alignment, power, and mass tradeoffs across two full flight-test campaigns at a UA-approved outdoor UAV test area.

Alabama Aerodynamics Experimental Research Operations (A.A.E.R.O.) **January 2025 – May 2026**

Co-Founder / CFD and Testing Lead

- Led design of a drone propeller with vortex inducers to evaluate aerodynamic performance.
- Simulated propeller thrust and acoustic characteristics using Computational Fluid Dynamics (CFD) in ANSYS and SolidWorks and compared results with real-world test data.
- Performed Finite Element Analysis (FEA) on designs to assess structural integrity and material performance.
- Applied Geometric Dimensioning and Tolerancing (GD&T) for precise manufacturing of components.
- Conducted static test stand experiments measuring thrust, RPM, and acoustics across multiple design iterations.
- Utilized a wind tunnel for aerodynamic data collection and model validation.

Alabama Rocketry Association — Liquids Team **October 2023 – May 2026**

Team Member

- Collaborated on designing components for a single-stage, liquid propellant rocket, focusing on fluid systems.
- Designed a test stand and created P&ID diagrams adhering to safety standards.
- Performed hydrostatic proof testing of flight hardware to demonstrate tank compliance with MIL-STD-810G/SMC-S-016 qualification requirements.
- Utilized a CNC lathe to manufacture a graphite nozzle within design tolerances.
- Assisted in setup and execution of multiple hot-fire tests for a LOX/Kerosene engine, including short-duration and duration tests at Tuscaloosa National Airport, analyzing performance data to identify areas for improvement.

Genesis Drone Simulation Project **August 2025 – May 2026**

Personal Project

- Built a high-fidelity quadcopter simulation in Python using the Genesis physics engine, modeling real-time aerodynamics, rigid-body dynamics, and gravity.
- Designed and tuned PID controllers for altitude, attitude, and position hold, achieving near-critical damping, sub-second settling times, and robust stability via anti-windup.
- Implemented manual and autonomous flight modes with differential motor RPM control and scaled the simulation to parallel multi-environment execution (>1000 steps/s on CUDA) with custom visualization tools.